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Paper Practice 3.3

Linearity and Continuity

Show all of your work in the space provided. Simplify before you decide — an equation can look nonlinear and simplify to a line.

1. Determine whether each function is linear or nonlinear. Simplify first. (Simplify First)

a. $y = 4x^2 - (2x)^2 + 3x - 5$

b. $y = 3x^3 - x^2 + 3x + 6$

c. $4x - (2y)^2 = 3$

2. Determine whether each function is linear or nonlinear. (Linear or Not)

a. $2x + 5y = 10$

b. $y = x^2 - 4$

c. $y = 7$

3. Simplify, then decide whether the function is linear or nonlinear. (Simplify First)

a. $y = 6x^2 - (3x)^2 + 3x^2 + 4x + 1$

4. Write each linear equation in standard form $ax + by = c$ with integer coefficients. (Standard Form)

a. $y = 3x - 8$

b. $y = -\frac{1}{2}x + 4$

5. Determine whether each situation is discrete or continuous. Explain. (Discrete or Continuous)

a. the number of tickets sold for a concert

b. the temperature of a room over one hour

c. the number of students riding a bus

6. The Bargain Book Barn sells novels on a sliding scale: the more you buy, the cheaper each one is. Let $f(x)$ be the price for x books. Is $f(x)$ discrete or continuous? Explain. (Discrete or Continuous)

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7. A function has domain {0, 1, 2, 3} and range {2, 5, 8, 11}. Is it discrete or continuous? Explain. (Discrete or Continuous)

8. The table shows a function. Is it linear? Justify your answer using the rate of change. (Rate of Change)

a. x: 1, 2, 3, 4 y: 3, 6, 9, 12

9. The table shows a function. Is it linear? Justify your answer. (Rate of Change)

a. x: 1, 2, 3, 4 y: 2, 4, 8, 16

10. Give the domain and the range of the discrete function {(−2, 1), (0, 3), (2, 5)}. (Domain & Range)

11. Error Analysis – Gus says $y = 4x^2 - (2x)^2 + 3x - 5$ is nonlinear because it contains x^2. Identify his mistake and give the correct answer. (Error Analysis)

12. Explain the difference between a discrete and a continuous function, then give a real-world example of each that is not used above. (Explain)